



Melamine foam/reduced graphene oxide supported form-stable phase change materials with simultaneous shape memory property and light-to-thermal energy storage capability

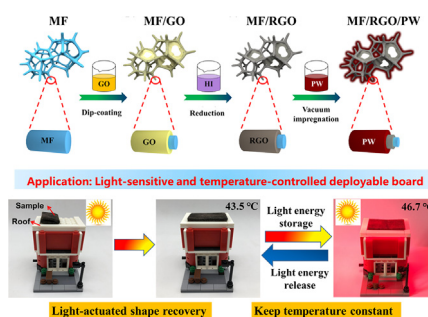
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HIGHLIGHTS

- Flexible phase change composites with simultaneous light-actuated shape variation property are prepared.
- Elastic MF prevents the liquid leakage and imparts the composites with shape recovery property.
- RGO network acts as a light absorption and heat transfer medium.
- A deployable board used for maintaining the temperature constancy of houses is explored.

GRAPHICAL ABSTRACT



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ABSTRACT

The advanced utilization of phase change materials (PCMs) is limited by the strong rigidity, liquid leakage and lack of photoabsorption ability. In this work, a novel form-stable PCM with solar-to-thermal energy storage ability and thermal/light-actuated shape memory property was prepared. Herein, paraffin wax (PW) served as a latent heat storage material and a switching phase for shape fixation, melamine foam (MF) acted as a supporting material to prevent the leakage and a permanent phase for shape recovery, and reduced graphene oxide (RGO) assembled on MF skeletons worked as a light absorption medium. Due to the light weight and abundant porosity of MF/RGO foam, the MF/RGO/PW PCMs possessed excellent encapsulation ability, high latent heat (144.8 J/g), good solar-to-thermal energy conversion and storage ability. Simultaneously, combining the elasticity of MF/RGO foam, the MF/RGO/PW PCMs exhibited thermal/light-sensitive shape memory property by triggering the phase transition of PW. The MF/RGO/PW PCMs showed excellent cyclic shape memory performance and thermal reliability. A light-sensitive and temperature-controlled board was further exploited, which could be installed on the roof of the house and maintained the temperature constancy. This study provides a strategy to prepare flexible, light-sensitive and form-stable PCMs via using the elastic MF/RGO foam as a supporting scaffold, and these PCMs have great potential in solar energy utilization and energy-saving buildings.

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1. Introduction

The increase of energy consumption, high cost of fossil fuels and environmental pollutions inspire researchers to develop renewable energy sources for a sustainable future. Thermal energy and solar energy, which are the most abundant and ubiquitous renewable sources, have the potential to replace fossil fuels [1]. Phase change materials (PCMs) are advanced energy storage materials that can store energy and release energy as the latent heat [2,3]. Therefore, PCMs have been applied in many fields, including energy-saving buildings, thermal management of electronic devices and solar energy harvesting, etc [4].

Paraffin wax (PW) is one kind of solid-liquid organic PCMs, which has been widely used because of its high latent heat, suitable phase change temperature and thermal stability [5]. However, the practical applications of paraffin inevitably face two major problems, i.e., the liquid leakage during phase change and the strong rigidity [6,7]. The former would give rise to the decrease of thermal energy storage efficiency and environment pollution. The latter would restrict the practical installation in confined electron device system. To solve the liquid leakage issue, form-stable PCMs are intensively investigated based on several methods such as microencapsulation, physical absorption and coaxial electrospinning [8–12]. Nevertheless, these conventional form-stable PCMs still face the installation difficulty that is caused by the strong rigidity of supporting scaffold [13]. Moreover, the rigidity of PCMs will give rise to easy brittle failure and poor surface contact with controlled devices [14]. Hence, it is of great significance to develop PCMs which have both excellent encapsulation ability and shape variation ability. Based on this consideration, the suitable *supporting material* should be carefully selected.

Inspired by previous studies of shape memory polymers (SMPs) that are capable of changing shape upon exposure to external stimuli, physical blending elastic polymers with paraffin may produce form-stable PCMs that also have the capability to realize shape variation [15–17]. Recently, melamine foam (MF) has received much attention due to its three-dimensional (3D) porous structure and high open porosity [18]. Besides, MF possesses an outstanding flexibility, this means MF can be deformed to arbitrary shapes and quickly recover to its original shape [19]. SMPs are usually consisted of switching phase and permanent phase [20]. The switching phase is responsible for fixing or unfixing the temporary shape, and the permanent shape works for memorizing the original shape. For MF/PW PCMs, MF serves as the permanent phase and PW serves as the switching phase. Specifically, MF provides the shape recovery force, while PW plays the role of shape deformation above the melting temperature (T_m) of PW and shape fixation below that point. Considering the porous structure and excellent elasticity of MF, the combination of PW and MF may impart MF/PW blends not only with good encapsulation ability but also with shape memory property [21].

In spite of the leakage and rigidity, the relatively low sunlight absorption ability of PCMs also inhibits their further applications in solar energy storage and conversion [22]. Various photothermal fillers including carbon nanotubes (CNTs), graphene nanoplatelets (GNPs), reduced graphene oxide (RGO), carbon aerogels and polydopamine have been incorporated into PCMs to improve the photoabsorption ability [23,33]. As one derivative of graphene, RGO has been demonstrated to be an ideal filler to endow form-stable PCMs with simultaneous solar-to-thermal energy conversion properties [34,35]. Interestingly, MF is often used as the template to build 3D interconnected RGO network [36–39]. The MF/RGO foams also possess open macropores and excellent flexibility, making them the ideal candidates for flexible supporting scaffold as well as solar energy conversion and storage.

In this work, MF was chosen as a template to prepare 3D continuous graphene architecture and introduced into PW via vacuum impregnation, aim at obtaining flexible PCM composites with high latent heat, good encapsulation effect, solar-to-thermal energy storage ability and thermal/light-actuated shape memory property. Herein, PW serves as a

latent heat storage material and a switching phase for shape fixation. The elastic MF works as a *supporting material* to prevent the liquid leakage and imparts the composites with shape recovery property. The RGO network acts as a light absorption and heat transfer medium. This novel MF/RGO/PW PCM composites are believed to have great promising applications in thermal/solar energy storage systems.

2. Materials and methods

2.1. Materials

Paraffin wax (PW) with a T_m of about 52 °C was purchased from Sinopharm Chemical Reagent Co., Ltd. (China). The density of PW is 0.82 g/cm³. Melamine foam (MF) was kindly supplied by Share Nano Technology Co., Ltd. (Xiamen, China). The density of MF is 0.025 g/cm³. The characterization of MF is shown in Fig. S1 (see *supporting information*). Graphene oxide (GO) was synthesized according to the Hummers method using natural graphite powder as the raw material, which was obtained from Heilong Co., Ltd. (Qingdao, China). The size of the GO sheet is about 5 ~ 10 μm and the thickness is about 1.0 nm.

2.2. Sample preparation

Preparation of MF/RGO foams: The MF/RGO foams were prepared by coating GO sheets onto the skeletons of MF (Fig. 1a) through a solution dip-coating method (Fig. 1b) and followed by chemical reduction in hydroiodic acid at 90 °C for 0.5 h (Fig. 1c). The RGO content in MF/RGO foams could be easily adjusted by changing the impregnation time and the concentration of GO suspension. In this study, MF with an ultralow density were selected as the template, and two types of MF/GO foams were fabricated: the MF/GO foam with one impregnation in GO suspension (labeled as MF/GO-1), and the MF/GO foam with two impregnations in GO suspension (labeled as MF/GO-2). The GO content in MF/GO foams was calculated by weighting the mass of the foam before and after the impregnation in GO suspension. The GO contents of MF/GO-1 and MF/GO-2 foams were about 5 wt% and 10 wt%, respectively. The density values of MF/RGO foams were calculated by the ratio of mass and volume. The MF/GO-1 and MF/GO-2 foams exhibited ultralow densities of about 0.026 and 0.028 g/cm³, respectively.

Preparation of MF/RGO/PW PCM composites: Firstly, 20 g PW was added into cyclohexane solution (47 mL) at 60 °C for 1 h. Then, MF/RGO foams (3 cm × 2 cm × 1 cm) were immersed into 50 mL PW/cyclohexane solution to absorb paraffin under the vacuum condition. After mixing for 10 min, the resulting materials were placed in a drying oven at 50 °C to remove the solvent and finally the MF/RGO/PW PCM composites were obtained (Fig. 1d). For comparison, the MF/PW PCM was also prepared using the same method. In this work, the weight ratio of PW to MF or MF/RGO was calculated as 20:1.

2.3. Characterizations and measurements

The morphologies of MF, MF/GO and MF/RGO were investigated by a scanning electron microscope (SEM, Fei Inspect, Netherlands) with an accelerating voltage of 5 kV. Fourier transform infrared spectroscopy (FTIR, Thermo Nicolet, USA) and Raman spectra (Horiba Evolution, France) with a laser of 532 nm were applied to investigate the structure of MF/RGO.

The morphologies of PCM composites were observed by SEM and optical microscope (OM, LEICA DM 2700P, Germany). FTIR and Wide-angle X-ray diffractometer (WAXD, Panalytical Empyrean) were used to study the structure of PCM composites. The X-Ray source was Ni-filtered Cu K α radiation which was conducted at 40 kV and 40 mA. A rheometer (TA DHR-1, USA) was employed to measure the shape stability of PCM composites. The samples were heated from 20 °C to 100 °C at 5 °C/min under a constant force of 0.3 N. During the heating process, the thickness change of the samples was recorded through the dynamic

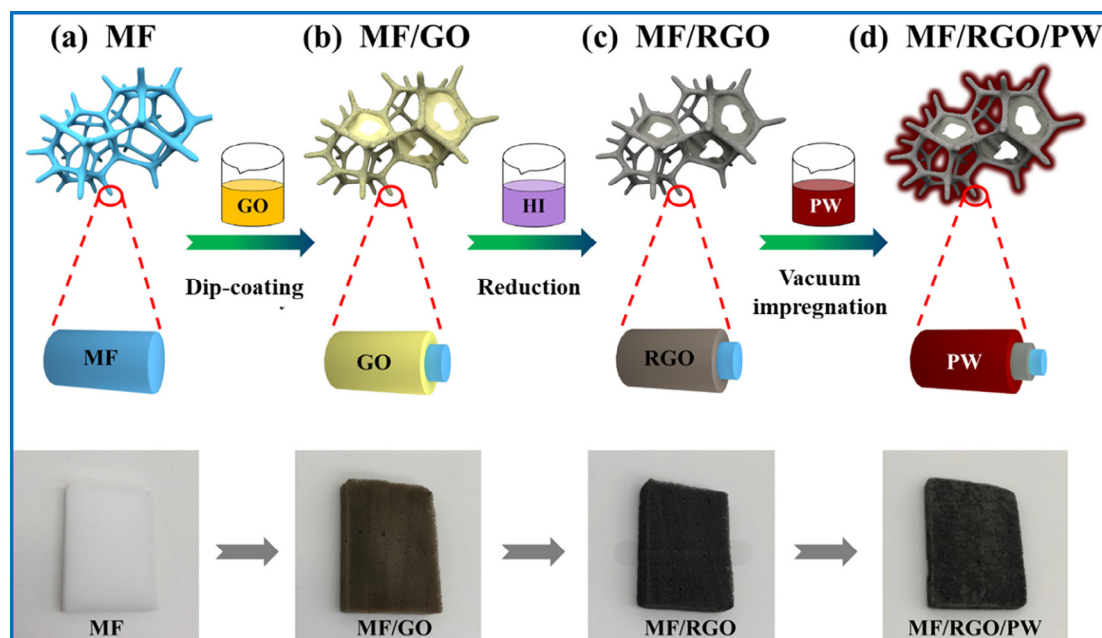


Fig. 1. Schematic representations showing the preparation procedure of MF/RGO/PW PCM composites. The images below showing the visual photographs of MF, MF/GO, MF/RGO and MF/RGO/PW, respectively.

reometer. Differential scanning calorimeter (DSC, Netzsch STA 449C, Germany) was conducted to measure the thermal-physical properties of PCM composites, which was set by heating the samples from 0 °C to 100 °C and cooling from 100 °C to 0 °C at 5 °C/min. The DSC measurements were operated in nitrogen atmosphere. The thermal energy storage capability of PCM composites was investigated through an infrared thermal imaging camera (FLIR T620, USA). The samples were placed on a heating stage at 80 °C. The light-to-thermal energy conversion behaviors of PCM composites were investigated by using a xenon lamp (HSX-F300, China) and the temperature variations were recorded with a digital thermocouple.

The shape memory properties of PCM composites were evaluated through a bend-recovery measurement. Firstly, the straight samples were heated to 60 °C and bent in half (180°). Secondly, the deformed sample ("U" shape) was cooled to 20 °C under a constant external force. Thirdly, the external force was removed and the fixation of temporary shape was recorded. Lastly, the samples were reheated to 60 °C and the shape recovery behavior was recorded by a digital camera. For light-actuated shape memory behavior, an infrared-light (IR) lamp (Philips, 150 W) was employed as the light source. The height of IR lamp was 35 cm and the power density was approximately 20 mW/cm².

3. Results and discussion

3.1. Morphology and structure of MF/RGO foams

The morphological evolution of MF/GO and MF/RGO foams was investigated by SEM. MF exhibits a porous and 3D interconnected network structure with macropores of about 100 μm (Fig. 2a). The overall morphology of MF/GO foam is similar to that of MF, but the skeletons of MF/GO foam show a rough and crinkled texture due to the existence of GO (Fig. 2b), indicating that GO sheets are successfully assembled around MF skeletons. After the reduction in hydroiodic acid, graphene sheets are still connected to each other (Fig. 2c). Notably, since MF/GO foams were prepared by a mild dip-coating method, some GO or RGO sheets were stacked on the skeletons of MF. Unlike the MF that is white in color, the as-obtained MF/GO and MF/RGO foams present uniform tawny and black appearance, respectively. In addition, the MF/RGO foam exhibits remarkable elasticity and flexibility. As

shown in Fig. 2d, the MF/RGO-1 foam can be compressed or twisted, and completely recovers their original shape without significant structural fatigue. Notably, the RGO sheets of MF/RGO foam cannot be stripped after hundreds of deformation cycles due to the hydrophobic and π - π interactions between RGO sheets and MF skeletons.[36] Based on above results, RGO foams with low density and high elasticity are successfully obtained using MF as a template. This super elasticity combined with the 3D graphene network would impart MF/RGO foams with unique functional properties.

The coating of GO on MF and the reduction of GO were confirmed by FTIR and Raman spectroscopy. Fig. 3a shows the FTIR spectra of MF, GO, MF/GO and MF/RGO. For neat MF, the peaks at 3427, 1620, 814 cm⁻¹ are the stretching vibration of N-H, stretching vibration of C = N and bending vibration of triazine ring, respectively.[40] After dipping in GO suspension, the peak at 3400 cm⁻¹ becomes broader, suggesting the existence of GO in MF. While for MF/RGO, the intensities of peaks corresponding to the oxygen functional groups including O-H groups and C=O groups decrease dramatically, indicating that most oxygen functional groups are reduced by hydroiodic acid. As shown in Fig. 3b, GO presents two strong peaks at 1336 cm⁻¹ and 1572 cm⁻¹, which are the D band and G band, respectively.[41] The Raman spectrum of MF/GO foam is similar to that of GO, validating that GO sheets are wrapped around MF skeletons. The ID/IG ratio of MF/RGO (1.10) is higher than that of MF/GO (0.97). This is due to more numerous but smaller sp² carbon domains are partially restored after the reduction of hydroiodic acid [42,43]. The results obtained from FTIR and Raman spectroscopy verify that GO sheets are coated onto MF skeletons and RGO sheets are successfully prepared through the reduction reaction of hydroiodic acid.

3.2. Morphology and structure of MF/RGO/PW PCM composites

MF and MF/RGO foams exhibit open macropores, which have the large capillary force to absorb PW into the inner pores. Then, MF and MF/RGO foams were immersed into the PW/cyclohexane solution to encapsulate PW. The morphologies of MF/PW and MF/RGO/PW PCM composites were characterized by SEM and optical microscope (OM). As shown in Fig. 4a and d, MF shows a 3D continuous network and the surface of MF skeleton is relatively smooth. Fig. 4b and e shows that the

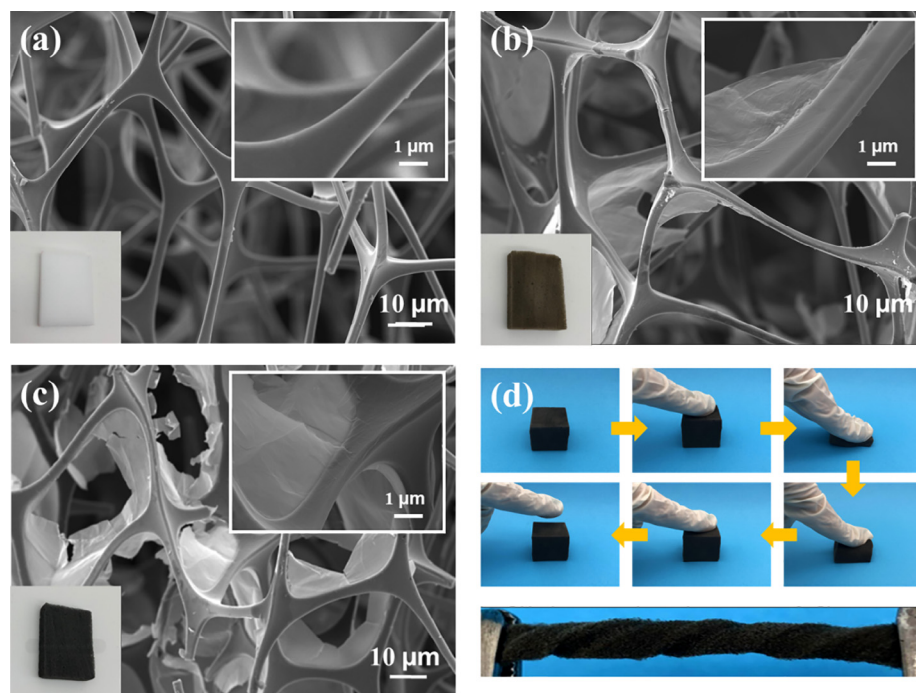


Fig. 2. SEM images and photographs of (a) MF, (b) MF/GO and (c) MF/RGO. (d) Compressing and twisting behaviors of MF/RGO-1 foam.

absorbed PW is presented along the skeleton or in the macropores of MF. For MF/RGO-1/PW composite PCM (Fig. 4c and f), it can be seen that both PW and RGO sheets exist in the pores of MF. Based on the SEM and OM observation, MF and MF/RGO foams can be well utilized as a porous matrix to absorb PW via vacuum impregnation.

To further investigate the structure of MF/RGO/PW PCM composites, FTIR and XRD analysis were carried out. Fig. 5a displays the FTIR spectra of MF, MF/RGO, PW, MF/PW and MF/RGO/PW. The characteristic peaks of PW at 2921, 1470, 1380, 719 cm^{-1} are the $-\text{CH}_2-$ asymmetrical stretching vibration, $-\text{CH}_2-$ deformation vibration, $-\text{CH}_3-$ deformation vibration and $-\text{CH}_2-$ rocking vibration, respectively [5]. For MF/PW and MF/RGO/PW PCM composites, only the characteristic peaks of PW is found and the absorption peaks of MF or RGO are almost invisible, suggesting that PW plays the dominant role. Besides, there are no new peaks form and no shift appears in the main absorption peaks of MF/RGO/PW PCM composites, indicating that no chemical reaction happens among MF, RGO and PW. MF/RGO foam is a polar material and PW ($\text{CH}_3-(\text{CH}_2)_n-\text{CH}_3$) is a non-polar material. The XRD patterns of PW, MF/PW, MF/RGO/PW are shown in Fig. 5b. Two diffraction peaks

at 21.2° and 23.6° corresponding to the (1 1 0) and (2 0 0) crystal planes of PW are observed. The diffraction peaks of MF/PW and MF/RGO/PW PCM composites are almost similar to the diffraction peaks of PW except little lower intensity caused by the decrease of the weight percentage of PW. Meanwhile, no obvious shift of diffraction peaks is observed, indicating the physical interpenetration between MF/RGO and PW.

3.3. Shape stability of MF/RGO/PW PCM composites

The shape stability of paraffin-based PCMs is crucial for preventing the leakage of melted PW. The shape stability of PW, MF/PW, and MF/RGO/PW was comparatively measured and the results are described in Fig. 6. As shown in Fig. 6a, neat PW cannot retain its shape when the temperature is above 40°C , and its dimension decreases seriously as the temperature further increases. When the temperature is about 65°C , the solid-liquid phase transition of PW occurs, leading to neat PW to leak completely. On the contrary, by impregnating PW into MF and MF/RGO foams, the dimensions of the resultant MF/PW and MF/RGO/PW

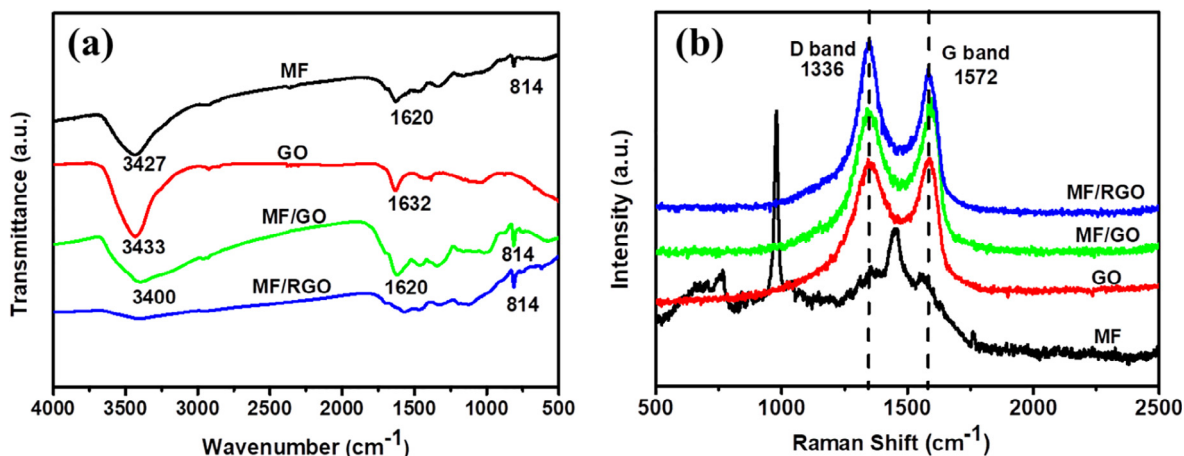


Fig. 3. (a) FTIR spectra and (b) Raman spectra of MF, GO, MF/GO and MF/RGO.

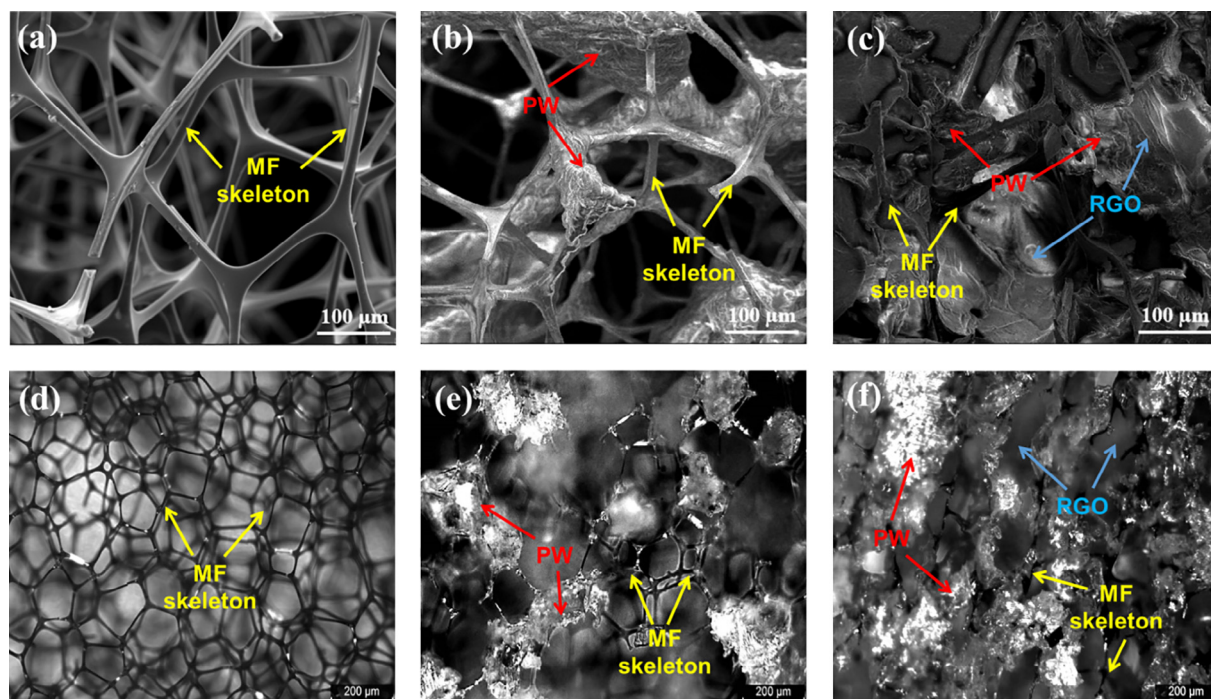


Fig. 4. (a-c) SEM images of MF, MF/PW and MF/RGO-1/PW. (d-f) OM images of MF, MF/PW and MF/RGO-1/PW.

PCM composites change little even at 80 °C. This result indicates that MF/PW and MF/RGO/PW PCM composites show remarkably improved shape stability because of the efficient support of MF framework. The leakage measurement was also performed via digital camera observation to directly evaluate the shape stability (Fig. 6b and Fig. S2). All of the samples can maintain their shape at 25 °C, which is below the T_m of PW. When heated to 80 °C, neat PW starts to melt with visible liquid leakage on the bottom. The red-dotted zone represents the leakage of melted PW. Due to the existence of MF or MF/RGO, the MF/PW and MF/RGO/PW PCM composites keep their stable shapes well during the whole heating process. Specially, even at a load of 20 g weight, the MF/RGO/PW PCM composites still maintain their original shape without obvious PW leakage, suggesting the MF/RGO framework is an ideal [supporting material](#) to improve the shape stability of PW. Therefore, the 3D interconnected network of MF/RGO encapsulates PW completely and imparts the MF/RGO/PW PCM composites with excellent shape stability.

3.4. Thermophysical properties of MF/RGO/PW PCM composites

Latent heat is very important for determining the thermal energy storage capacity of PCMs. The DSC analysis was then conducted to investigate the phase change properties such as enthalpies and phase transition temperatures. The DSC melting and cooling curves of PW, MF/PW and MF/RGO/PW are shown in Fig. 7. The thermal parameters including T_m , crystallization temperature (T_c), melting enthalpy (ΔH_m) and crystallization enthalpy (ΔH_c) are summarized in Table 1. For neat PW, two main melting peaks are observed, which are assigned to the crystal-rotator transition and rotator-liquid transition of PW crystals. [15] Compared with neat PW, the T_m s are not influenced obviously with the addition of MF or MF/RGO (Fig. 7a). The ΔH_m of MF/PW and MF/RGO/PW PCMs decreases in a certain extent due to the decrease of the weight percentage of PW. Here, the weight ratio of PW to MF or MF/RGO is 20:1. A slight decrease in the ΔH_m is observed for a MF/PW with 141.6 J/g and a MF/RGO-1/PW with 136.3 J/g compared with neat PW with 165 J/g. The thermal energy storage capability of MF/PW, MF/RGO-1/PW and MF/RGO-2/PW PCMs reaches as high as 85.8%, 82.6%

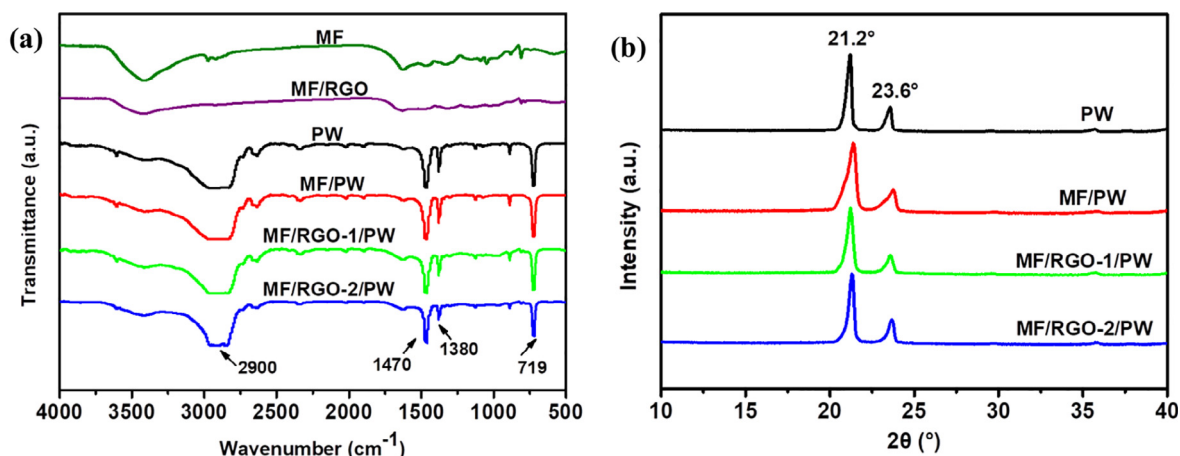


Fig. 5. (a) FTIR spectra for MF, MF/RGO, PW, MF/PW and MF/RGO/PW. (b) XRD patterns of PW, MF/PW, MF/RGO/PW.

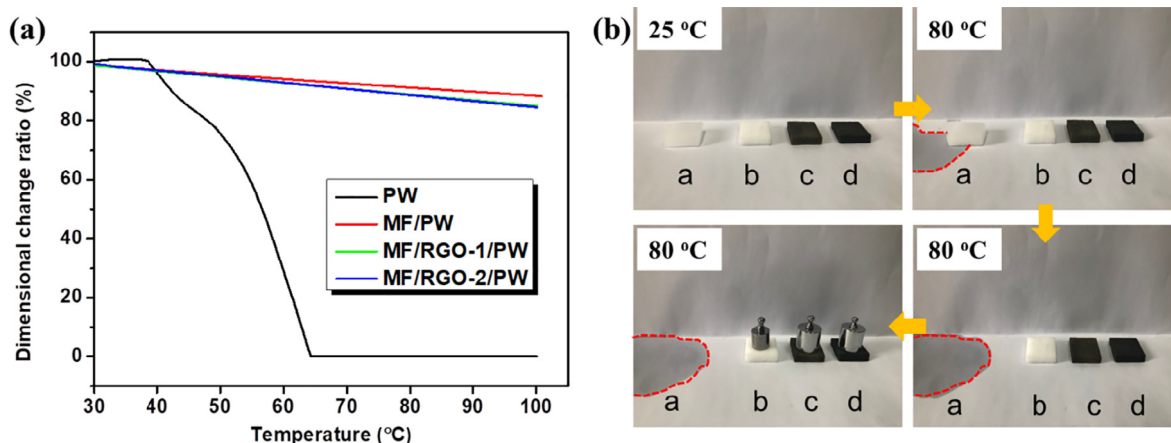


Fig. 6. (a) Thermomechanical analyses and (b) digital photographs showing the shape stability of PW, MF/PW, and MF/RGO/PW at 25 °C and 80 °C. Samples from left to right are a) PW, b) MF/PW, c) MF/RGO-1/PW and d) MF/RGO-2/PW, respectively. The red-dotted zone represents the leakage of melted PW.

and 87.8% of neat PW, respectively. Similar phenomenon is also found in the DSC cooling thermograms (Fig. 7b). Due to the extremely low density of MF/GO foam, the thermal storage capability of MF/RGO/PW PCMs is still higher than previously reported PCMs encapsulated by polymers [16,44–46].

Since the MF/RGO/PW PCMs possess large latent heat and good encapsulation effect, the thermal energy storage and heat transfer capabilities were further investigated. The MF/PW and MF/RGO-1/PW samples were placed on a heating stage with a constant temperature of

Table 1

Thermal characteristics of PW, MF/PW and MF/RGO/PW.

Sample	$T_m/(^{\circ}\text{C})$	$T_c/(^{\circ}\text{C})$	$\Delta H_m/(\text{J g}^{-1})$	$\Delta H_c/(\text{J g}^{-1})$
PW	55.7	47.3	165.0	165.7
MF/PW	55.8	44.6	141.6	149.3
MF/RGO-1/PW	56.8	44.2	136.3	150.9
MF/RGO-2/PW	57.8	44.1	144.8	154.1

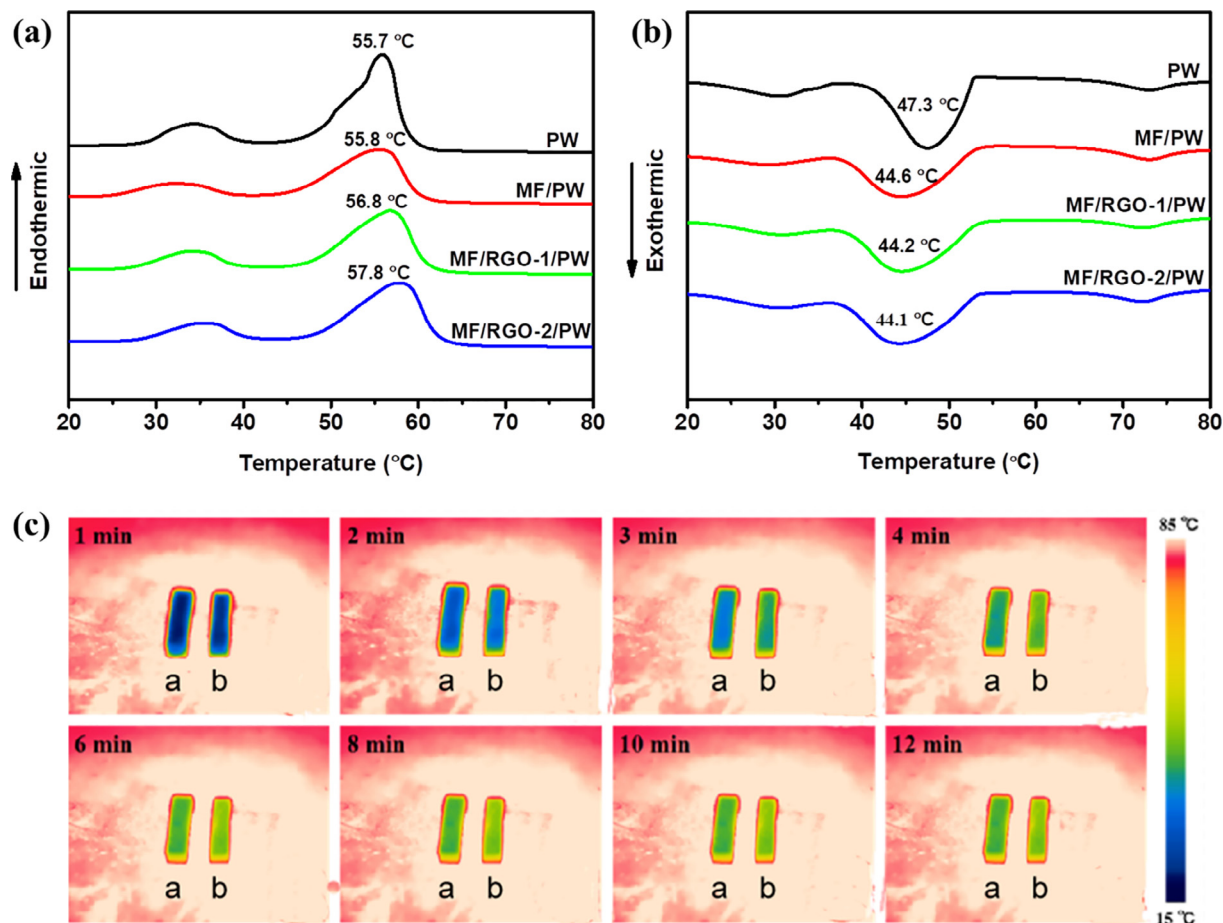


Fig. 7. (a) DSC heating curves and (b) cooling curves of PW, MF/PW and MF/RGO/PW. (c) Infrared thermal images showing the thermal energy storage and heat transfer behaviors of a) MF/PW and b) MF/RGO-1/PW.

80 °C, and an infrared thermal imager was employed to record the temperature response during heating. Fig. 7c illustrates the temperature distribution images of MF/PW and MF/RGO-1/PW PCMs at different time intervals. It can be clearly seen that the temperature of MF/RGO-1/PW increases faster than that of MF/PW, which indicates that MF/RGO-1/PW has higher thermal conductivity and faster heat transfer rate. The MF/RGO-1/PW exhibits a thermal conductivity of $0.096 \text{ W m}^{-1} \text{ K}^{-1}$, which is enhanced by 54.8% compared with MF/PW. The 3D RGO network assembled on the MF skeleton provides an efficient thermal transport pathway, leading to the enhancement of thermal conductivity and shortening the time needed for thermal storage. [38] Besides, both MF/PW and MF/RGO-1/PW PCMs present a temperature plateau for about 6 min, which is originated from the solid-liquid phase transition of PW. In this condition, the MF/RGO-1/PW PCM absorb thermal energy during phase change and maintain the temperature constant around the T_m of PW. The MF/RGO/PW PCMs possess excellent thermal storage capability and enhanced thermal transport property than MF/PW, thus showing promising application in the thermal preservation systems.

3.5. Light-to-thermal energy storage property of MF/RGO/PW PCM composites

Nowadays, the utilization of sustainable energy becomes more important, and thus the exploitation of solar energy is attracting great attention. The light-to-thermal energy conversion and storage behavior of MF/PW and MF/RGO/PW PCMs were investigated by placing the samples under a xenon lamp with an intensity of 100 mW/cm^2 , and the temperature variation was recorded using a thermocouple, as shown in Fig. 8a. The temperature evolution curves of the samples exposed to solar irradiation is displayed in Fig. 8b. Compared with MF/PW, the temperature of MF/RGO/PW rises more rapidly, and MF/RGO-2/PW exhibits higher temperature than MF/RGO-1/PW. This phenomenon indicates that RGO sheets are effective photon captors and molecular heaters, and can absorb solar energy, convert solar energy into heat energy, and store that energy in PCM composites. Obviously, a higher loading of RGO can absorb more light energy and convert more thermal energy in unit time. As the exposure time increases, a temperature growth plateau at about 50 °C emerges and maintains constant for 2 min, which corresponds to the solid-liquid phase transition of PW. Once the temperature of MF/RGO/PW PCMs reaches the T_m of PW, the transferred thermal energy is stored in the form of latent heat. After removing the simulant light source, the temperature of MF/RGO/PW PCMs presents a freezing plateau between about 54 °C and 49 °C, indicating that the crystallization of PW releases the stored thermal

energy and then the thermal energy is transferred to the whole sample during the solidification process. On the contrary, no visible melting or cooling plateaus are found for MF/PW, which demonstrates that MF/PW exhibits weak photo-absorption ability and the phase change does not happen during the light irradiation process.

3.6. Shape memory behavior of MF/RGO/PW PCM composites

In addition to the encapsulate property and latent heat, the strong rigidity of PCMs may cause easy brittle failure and installation difficulty in practical applications [6]. Flexible PCMs which have shape variation capability are expected to solve this problem. Unlike other studies where the flexibility of PCMs is restricted due to the rigidity of supporting scaffolds, the super elasticity of MF/RGO foams may allow the MF/RGO/PW PCMs being capable of reversible shape deformation and recovery. In order to confirm our assumption, the thermal-actuated shape memory behaviors of MF/PW and MF/RGO/PW PCMs were further investigated. All the samples were cut into strips and then bent in half (180°) at 60 °C. As shown in Fig. 9a, after cooling to 20 °C and releasing load, the temporary bended shape (“U” shape) can be well fixed, indicating that MF/PW and MF/RGO/PW PCMs have good shape fixing properties. When the bended samples are placed on the heating stage at 80 °C, they are capable to recover their original straight shape. Notably, the shape recovery time of MF/RGO-1/PW (220 s) and MF/RGO-2/PW (190 s) is lower than that of MF/PW (260 s), suggesting that RGO can accelerate the shape recovery speed. This is due to the 3D continuous RGO network provides the thermal conduction pathway and thus promotes the heat transfer throughout the materials. In this study, MF and MF/RGO foams have an outstanding elasticity, but they cannot fix the deformed shape without external force at 20 °C. From above SEM observation, PW is fulfilled in the pores of MF or MF/RGO foams. The T_c of PW is about 47 °C, and the PW crystals are responsible for fixing the temporary bended shape at 20 °C. Upon heating above the T_m of PW, PW crystals melt and the PCM composites are transformed into flexible block. Consequently, the resilience of MF or MF/RGO provides the shape recovery force, leading to the shape recovery from a temporary bended shape to an original straight shape.

Light-actuated SMPs are also attractive due to their convenient manipulation and remote controllability. RGO sheets are capable to exhibit light absorption and energy conversion because of their good photothermal effect [47]. The light-actuated shape memory behavior of MF/PW and MF/RGO/PW PCMs is shown in Fig. 9b. The MF/PW PCM has low light absorption and it cannot be driven by the IR irradiation. With the addition of RGO, the MF/RGO/PW PCMs show rapid shape recovery effect and the bended shape can be fully recovered to the

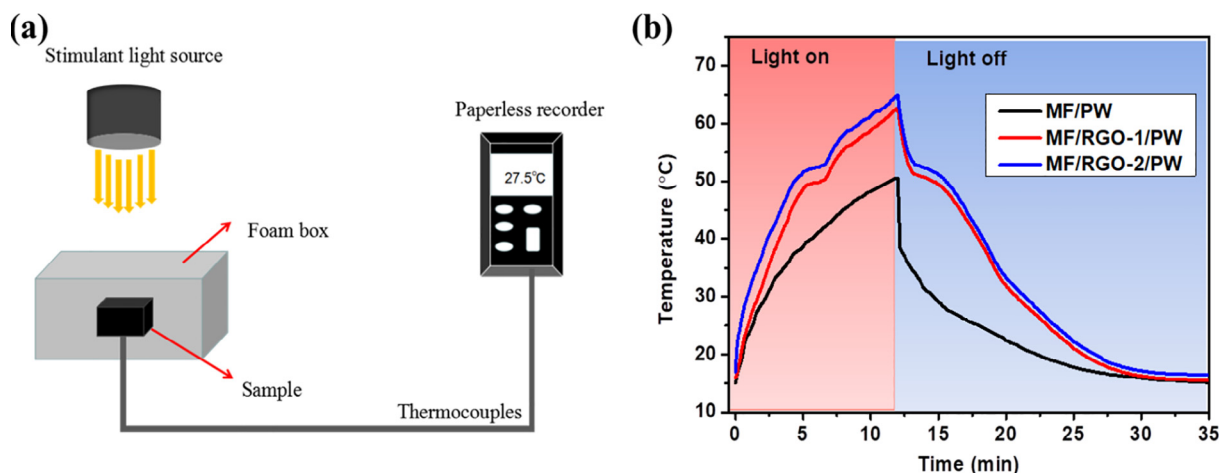


Fig. 8. (a) Schematic diagram of the light-to-thermal energy conversion and storage testing system. (b) Temperature evolution curves of MF/PW and MF/RGO/PW PCMs under stimulant solar irradiation.

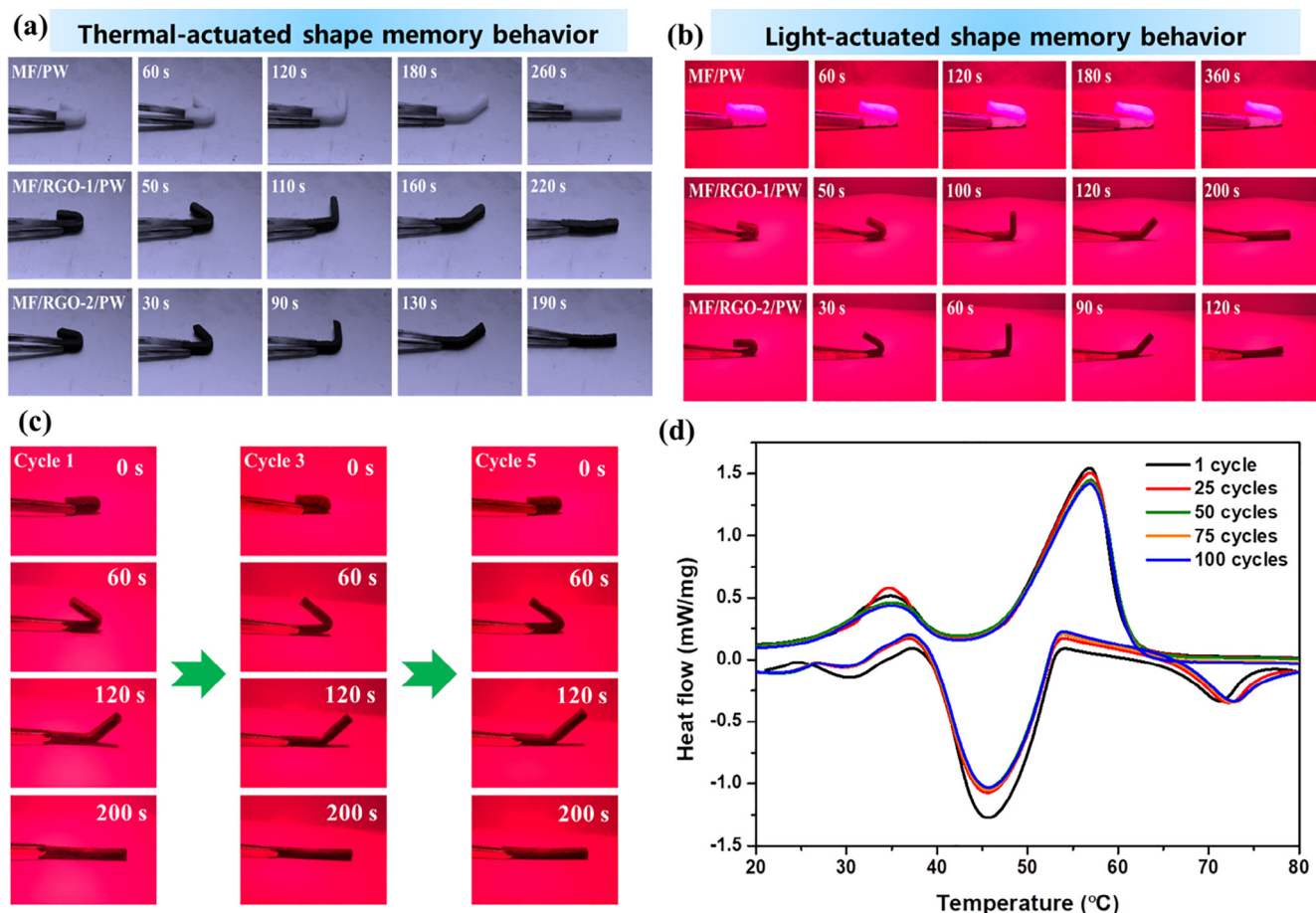


Fig. 9. (a) Thermal-actuated shape recovery behaviors and (b) Light-actuated shape recovery behaviors of MF/PW and MF/RGO/PW PCMs. (c) The light-actuated shape recovery behaviors of MF/RGO-1/PW after 1, 3 and 5 cycles. (d) DSC heating and cooling curves of MF/RGO-1/PW after 100 cycles.

straight shape within a short time. The more RGO content, the faster shape recovery speed is. In this case, RGO sheets work as light-absorber and heat source, thus converting light energy into thermal energy and making the temperature of MF/RGO/PW PCMs increase.

Reliability and stability are also two main issues for flexible PCMs in terms of long-time usage. Take the MF/RGO-1/PW PCM as an example, the light-actuated shape memory cyclic performance and the thermal stability were investigated. Fig. 9c shows the light-actuated shape recovery behaviors of MF/RGO-1/PW PCM after 1, 3 and 5 cycles. No changes in the shape recovery behavior and speed are observed after 5 cycles, which suggests that the shape memory property of MF/RGO-1/PW PCM is quite stable. Thermal stability was evaluated by performing DSC thermal cycling measurement. The DSC thermograms of MF/RGO-1/PW PCM after 100 melting and cooling cycles are shown in Fig. 9d. It's noteworthy that the phase change temperatures and latent heat almost change negligibly. Besides, the FTIR spectra (Fig. S3) show that no new peak appears or peak disappears after 100 cycles. Therefore, the MF/RGO/PW PCMs show excellent cyclic light-actuated shape memory property and thermal stability, indicating the long-term usability for practical applications.

3.7. Potential application in the field of energy-saving buildings

The prepared MF/RGO/PW PCM composites with simultaneous functions of light/thermal energy storage and shape variation could show advantages in some specific areas. One important potential application is the thermal management of buildings. As shown in Fig. 10a, the MF/RGO/PW PCM composites can be utilized as a light-sensitive deployable board and maintain the temperature constancy of the roof of

the house. On one hand, the shape fixing and shape recovery properties endow the MF/RGO-1/PW PCM composites with good installation ability. The board can be bent into a smaller shape so that it is easily installed on the roof of the house. Here, IR lamp is used as the simulated light source. In the day time (under light irradiation), RGO absorbs solar energy in near infrared region and converts the light energy into thermal energy. When the temperature of the board increases to the T_m of PW, the temporary bended shape starts to recover to its original flat shape, and then the board gradually covers the whole roof (Fig. 10b). On the other hand, high latent heat imparts the MF/RGO-1/PW PCM composites with good light-to-thermal energy storage capability and thus the board can keep the roof at a constant temperature. During continuous light irradiation, the surface temperature of MF/RGO-1/PW PCM composites maintains constant around the T_m of PW (Fig. 10c). In this circumstances, the phase transition and the light-to-thermal energy storage take place, accompanying with a small variation of temperature when MF/RGO-1/PW absorbs solar irradiation, converts light energy into heat energy, and stores heat energy. After removing the light irradiation, the temperature of the board drops very slowly, suggesting a small temperature fluctuation (Fig. 10d). This is ascribed to PW solidifies and a large amount of stored heat energy is released in the cooling phase change process. Through the process of light-to-thermal energy storage and release, the board can maintain the temperature of the roof at a relatively constant level for a long time. Therefore, the light-sensitive and temperature-controlled MF/RGO/PW PCM composites can be easily installed in some confined areas of houses, and absorbs solar energy during the day and it slowly releases the heat at night.

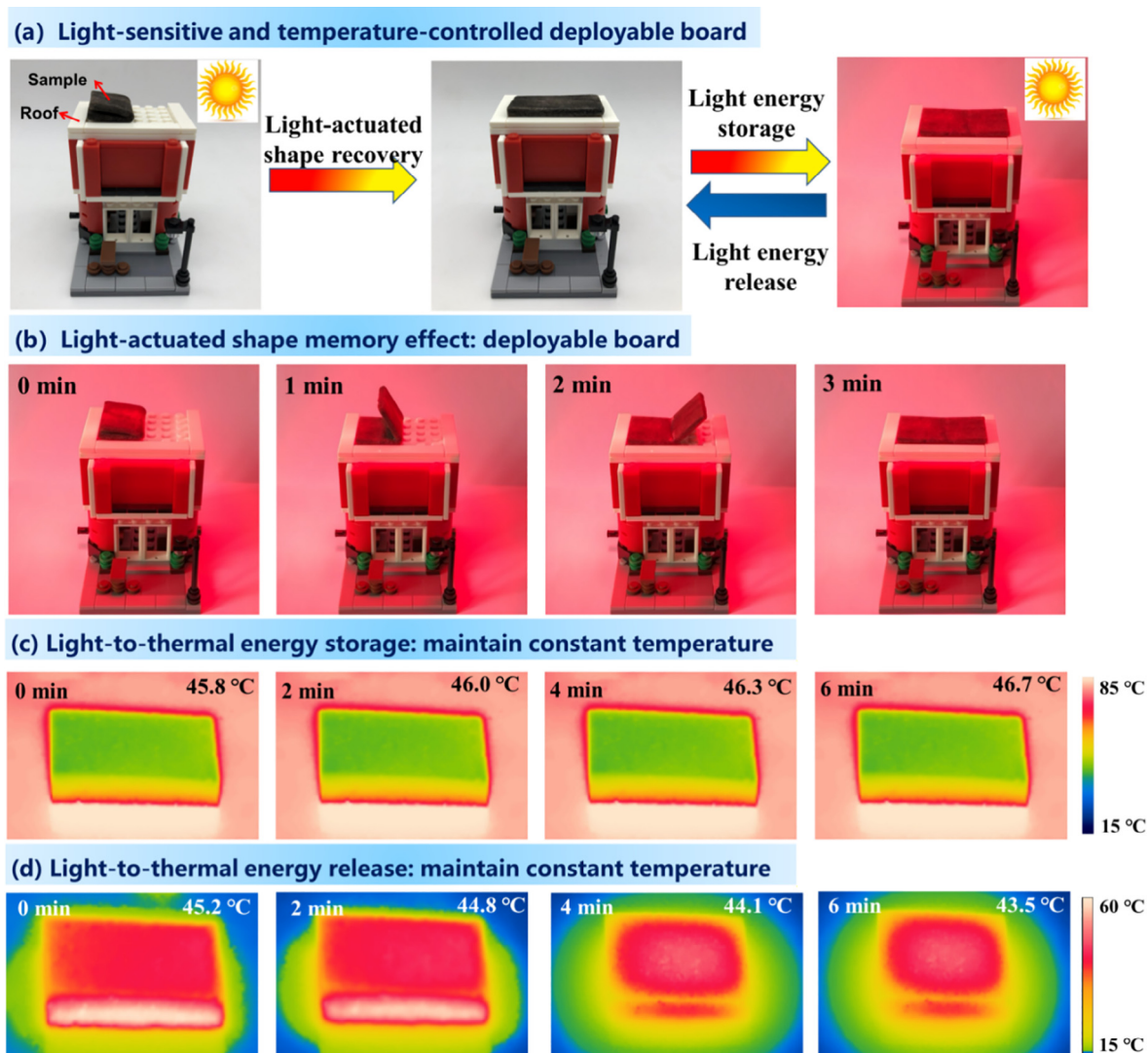


Fig. 10. (a) The MF/RGO-1/PW PCM composite used as a light-sensitive and temperature-controlled deployable board to maintain the temperature of roof. (b) Digital photographs show the shape recovery from a bended shape to a flat shape under IR irradiation. Infrared photographs show the constant surface temperature of board (c) under light irradiation and (d) removing light irradiation.

4. Conclusions

In this work, a novel flexible MF/RGO/PW PCM composite with high latent heat, good encapsulation ability, solar-to-thermal energy storage ability and thermal/light-actuated shape memory property is successfully prepared. MF is selected as a template to build 3D continuous RGO network and then MF/RGO foam is introduced into PW through vacuum impregnation. Herein, PW serves as a latent heat storage material, MF acts as a supporting material and RGO works as a light absorption medium. Due to the light density and high porosity of MF/RGO foam, PW can be loaded into MF/RGO foam with a high loading of up to approximately 2000 wt%. The MF/RGO/PW PCM composites show high latent heat (144.8 J/g) and good encapsulation effect. The RGO network imparts the MF/RGO/PW PCM composites with enhanced thermal transport property and solar-to-thermal storage capability. More interestingly, by triggering the phase transition of PW upon the thermal or light stimuli, the MF/RGO/PW PCM composites exhibit excellent shape fixing and recovery property. In addition, the PCM composites possess stable shape memory cyclic property and thermal reliability. One potential application of deployable board used for maintaining the temperature constancy of houses is further demonstrated. The superelastic MF/RGO foam supported PCM composites

have great potential in advanced applications such as solar energy storage and energy-saving buildings.

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Appendix A. Supplementary data

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References

- [1] J. Yang, L.S. Tang, L. Bai, R.Y. Bao, Z.Y. Liu, B.H. Xie, M.B. Yang, W. Yang, High-performance composite phase change materials for energy conversion based on macroscopically three-dimensional structural materials, *Mater. Horiz.* 6 (2019) 250–273.
- [2] W. Aftab, X. Huang, W. Wu, Z. Liang, A. Mahmood, R. Zou, Nanoconfined phase

- change materials for thermal energy applications, *Energ. Environ. Sci.* 11 (2018) 1392–1424.
- [3] H.Y. Gao, J.J. Wang, X. Chen, G. Wang, X.B. Huang, A. Li, W.J. Dong, Nanoconfinement effects on thermal properties of nanoporous shape-stabilized composite PCMs: a review, *Nano Energy* 53 (2018) 769–797.
 - [4] X. Wei, F. Xue, X.D. Qi, J.H. Yang, Z.W. Zhou, Y.P. Yuan, Y. Wang, Photo- and electro-responsive phase change materials based on highly anisotropic microcrystalline cellulose/graphene nanoplatelet structure, *Appl. Energ.* 236 (2019) 70–80.
 - [5] Q. Lian, Y. Li, A.A.S. Sayyed, J. Cheng, J. Zhang, Facile strategy in designing epoxy/paraffin multiple phase change materials for thermal energy storage applications, *ACS Sustain. Chem. Eng.* 6 (2018) 3375–3384.
 - [6] W.W. Li, W.L. Cheng, B. Xie, N. Liu, L.S. Zhang, Thermal sensitive flexible phase change materials with high thermal conductivity for thermal energy storage, *Energ. Convers. Manage.* 149 (2017) 1–12.
 - [7] W. Aftab, A. Mahmood, W. Guo, M. Yousaf, H. Tabassum, X. Huang, Z. Liang, A. Cao, R. Zou, Polyurethane-based flexible and conductive phase change composites for energy conversion and storage, *Energy Storage Mater.* (2018), <https://doi.org/10.1016/j.ensm.2018.10.014>.
 - [8] X.B. Huang, X. Chen, A. Li, D. Atinafu, H.Y. Gao, W.J. Dong, G. Wang, Shape-stabilized phase change materials based on porous supports for thermal energy storage applications, *Chem. Eng. J.* 356 (2019) 641–661.
 - [9] X. Xiao, P. Zhang, M. Li, Preparation and thermal characterization of paraffin/metal foam composite phase change material, *Appl. Energ.* 112 (2013) 1357–1366.
 - [10] Q. Zhang, Z. He, X. Fang, X. Zhang, Z. Zhang, Experimental and numerical investigations on a flexible paraffin/fiber composite phase change material for thermal therapy mask, *Energy Storage Mater.* 6 (2017) 36–45.
 - [11] Y. Lu, X. Xiao, Y. Zhan, C. Huan, S. Qi, H. Cheng, G. Xu, Core-sheath paraffin-wax-loaded nanofibers by electrospinning for heat storage, *ACS Appl. Mater. Inter.* 10 (2018) 12759–12767.
 - [12] Y. Lu, X.D. Xiao, J. Fu, C.M. Huan, S. Qi, Y.J. Zhan, Y.Q. Zhu, G. Xu, Novel smart textile with phase change materials encapsulated core-sheath structure fabricated by coaxial electrospinning, *Chem. Eng. J.* 355 (2019) 532–539.
 - [13] Q.R. Sun, H.Q. Zhang, J.J. Xue, X.P. Yu, Y.P. Yuan, X.L. Cao, Flexible phase change materials for thermal storage and temperature control, *Chem. Eng. J.* 353 (2018) 920–929.
 - [14] W. Wu, W. Wu, S. Wang, Form-stable and thermally induced flexible composite phase change material for thermal energy storage and thermal management applications, *Appl. Energ.* 236 (2019) 10–21.
 - [15] Q. Zhang, Y. Zhao, J. Feng, Systematic investigation on shape stability of high-efficiency SEBS/paraffin form-stable phase change materials, *Sol. Energ. Mat. Sol. C* 118 (2013) 54–60.
 - [16] Q.L. Zhang, J.C. Feng, Difunctional olefin block copolymer/paraffin form-stable phase change materials with simultaneous shape memory property, *Sol. Energ. Mat. Sol. C* 117 (2013) 259–266.
 - [17] G. Li, H. Zhang, D. Fortin, W. Fan, H. Xia, Y. Zhao, A composite material with room temperature shape processability and optical repair, *J. Mater. Chem. C* 4 (2016) 5932–5939.
 - [18] Y. Feng, J. Yao, Design of melamine sponge-based three-dimensional porous materials toward applications, *Ind. Eng. Chem. Res.* 57 (2018) 7322–7330.
 - [19] C. Ruan, K. Ai, X. Li, L. Lu, A superhydrophobic sponge with excellent absorbency and flame retardancy, *Angew. Chem. Int. Ed.* 53 (2014) 5556–5560.
 - [20] T. Xie, Recent advances in polymer shape memory, *Polymer* 52 (2011) 4985–5000.
 - [21] J.H. Jing, H.Y. Wu, Y.W. Shao, X.D. Qi, J.H. Yang, Y. Wang, Melamine foam-supported form-stable phase change materials with simultaneous thermal energy storage and shape memory properties for thermal management of electronic devices, *ACS Appl. Mater. Inter.* 11 (2019) 19252–19259.
 - [22] S. Tahan Latibari, S.M. Sadrameli, Carbon based material included-shaped stabilized phase change materials for sunlight-driven energy conversion and storage: An extensive review, *Sol. Energy* 170 (2018) 1130–1161.
 - [23] L. Chen, R. Zou, W. Xia, Z. Liu, Y. Shang, J. Zhu, Y. Wang, J. Lin, D. Xia, A. Cao, Electro and photodriven phase change composites based on wax-infiltrated carbon nanotube sponges, *ACS Nano* 6 (2012) 10884–10892.
 - [24] R. Chen, R. Yao, W. Xia, R. Zou, Electro/photo to heat conversion system based on polyurethane embedded graphite foam, *Appl. Energ.* 152 (2015) 183–188.
 - [25] Y. Wang, B. Tang, S. Zhang, Single-walled carbon nanotube/phase change material composites: sunlight-driven, reversible, form-stable phase transitions for solar thermal energy storage, *Adv. Funct. Mater.* 23 (2013) 4354–4360.
 - [26] J. Yang, E. Zhang, X. Li, Y. Zhang, J. Qu, Z.Z. Yu, Cellulose/graphene aerogel supported phase change composites with high thermal conductivity and good shape stability for thermal energy storage, *Carbon* 98 (2016) 50–57.
 - [27] W.T. Wang, B.T. Tang, B.Z. Ju, Z.M. Gao, J.H. Xiu, S.F. Zhang, Fe₃O₄-functionalized graphene nanosheet embedded phase change material composites: efficient magnetic- and sunlight-driven energy conversion and storage, *J. Mater. Chem. A* 5 (2017) 958–968.
 - [28] G. Qi, J. Yang, R. Bao, D. Xia, M. Cao, W. Yang, M. Yang, D. Wei, Hierarchical graphene foam-based phase change materials with enhanced thermal conductivity and shape stability for efficient solar-to-thermal energy conversion and storage, *Nano Res.* 10 (2016) 802–813.
 - [29] J. Yang, G.Q. Qi, Y. Liu, R.Y. Bao, Z.Y. Liu, W. Yang, B.H. Xie, M.B. Yang, Hybrid graphene aerogels/phase change material composites: Thermal conductivity, shape-stabilization and light-to-thermal energy storage, *Carbon* 100 (2016) 693–702.
 - [30] J. Yan, G.Q. Qi, R.Y. Bao, K.Y. Yi, M.L. Li, L. Peng, Z. Cai, M.B. Yang, D.C. Wei, W. Yang, Hybridizing graphene aerogel into three-dimensional graphene foam for high-performance composite phase change materials, *Energy Storage Mater.* 13 (2018) 88–95.
 - [31] J. Yang, G.Q. Qi, L.S. Tang, R.Y. Bao, L. Bai, Z.Y. Liu, W. Yang, B.H. Xie, M.B. Yang, Novel photodriven composite phase change materials with bioinspired modification of BN for solar-thermal energy conversion and storage, *J. Mater. Chem. A* 4 (2016) 9625–9634.
 - [32] Y. Li, Y.A. Samad, K. Polychronopoulou, S.M. Alhassan, K. Liao, From biomass to high performance solar-thermal and electric-thermal energy conversion and storage materials, *J. Mater. Chem. A* 2 (2014) 7759–7765.
 - [33] Y. Zhang, G.G. Gurzadyan, M.M. Umair, W. Wang, R. Lu, S. Zhang, B. Tang, Ultrafast and efficient photothermal conversion for sunlight-driven thermal-electric system, *Chem. Eng. J.* 344 (2018) 402–409.
 - [34] L.S. Tang, J. Yang, R.Y. Bao, Z.Y. Liu, B.H. Xie, M.B. Yang, W. Yang, Polyethylene glycol/graphene oxide aerogel shape-stabilized phase change materials for photo-to-thermal energy conversion and storage via tuning the oxidation degree of graphene oxide, *Energ. Convers. Manage.* 146 (2017) 253–264.
 - [35] Y. Zhou, X. Liu, D. Sheng, C. Lin, F. Ji, L. Dong, S. Xu, H. Wu, Y. Yang, Polyurethane-based solid-solid phase change materials with in situ reduced graphene oxide for light-thermal energy conversion and storage, *Chem. Eng. J.* 338 (2018) 117–125.
 - [36] C. Wu, X. Huang, X. Wu, R. Qian, P. Jiang, Mechanically flexible and multifunctional polymer-based graphene foams for elastic conductors and oil-water separators, *Adv. Mater.* 25 (2013) 5658–5662.
 - [37] C.W. Li, D.G. Jiang, H. Liang, B.B. Huo, C.Y. Liu, W.R. Yang, J.Q. Liu, Superelastic and arbitrary-shaped graphene aerogels with sacrificial skeleton of melamine foam for varied applications, *Adv. Funct. Mater.* 28 (2018) 1704674.
 - [38] M. Qin, Y. Xu, R. Cao, W. Feng, L. Chen, Efficiently controlling the 3D thermal conductivity of a polymer nanocomposite via a hyperelastic double-continuous network of graphene and sponge, *Adv. Funct. Mater.* 28 (2018) 1805053.
 - [39] H.B. Yao, J. Ge, C.F. Wang, X. Wang, W. Hu, Z.J. Zheng, Y. Ni, S.H. Yu, A flexible and highly pressure-sensitive graphene-polyurethane sponge based on fractured microstructure design, *Adv. Mater.* 25 (2013) 6692–6698.
 - [40] A. Stolz, S. Le Floch, L. Reinert, S.M.M. Ramos, J. Tualion-Combes, Y. Soneda, P. Chaudet, D. Baillis, N. Blanchard, L. Duclaux, A. San-Miguel, Melamine-derived carbon sponges for oil-water separation, *Carbon* 107 (2016) 198–208.
 - [41] J. Zhang, H. Yang, G. Shen, P. Cheng, J. Zhang, S. Guo, Reduction of graphene oxide via L-ascorbic acid, *Chem. Commun.* 46 (2010) 1112–1114.
 - [42] S. Stankovich, D.A. Dikin, R.D. Piner, K.A. Kohlhaas, A. Kleinhammes, Y. Jia, Y. Wu, S.T. Nguyen, R.S. Ruoff, Synthesis of graphene-based nanosheets via chemical reduction of exfoliated graphite oxide, *Carbon* 45 (2007) 1558–1565.
 - [43] B. Shen, Y. Li, W. Zhai, W. Zheng, Compressible graphene-coated polymer foams with ultralow density for adjustable electromagnetic interference (EMI) shielding, *ACS Appl. Mater. Inter.* 8 (2016) 8050–8057.
 - [44] M. Mu, P.A.M. Basheer, W. Sha, Y. Bai, T. McNally, Shape stabilised phase change materials based on a high melt viscosity HDPE and paraffin waxes, *Appl. Energ.* 162 (2016) 68–82.
 - [45] F. Chen, M. Wolcott, Polyethylene/paraffin binary composites for phase change material energy storage in building: a morphology, thermal properties, and paraffin leakage study, *Sol. Energ. Mat. Sol. C* 137 (2015) 79–85.
 - [46] B. Tian, W. Yang, L. Luo, J. Wang, K. Zhang, J. Fan, J. Wu, T. Xing, Synergistic enhancement of thermal conductivity for expanded graphite and carbon fiber in paraffin/EVA form-stable phase change materials, *Sol. Energy* 127 (2016) 48–55.
 - [47] J.J. Liang, Y.F. Xu, Y. Huang, L. Zhang, Y. Wang, Y.F. Ma, F.F. Li, T.Y. Guo, Y.S. Chen, Infrared-triggered actuators from graphene-based nanocomposites, *J. Phys. Chem. C* 113 (2009) 9921–9927.